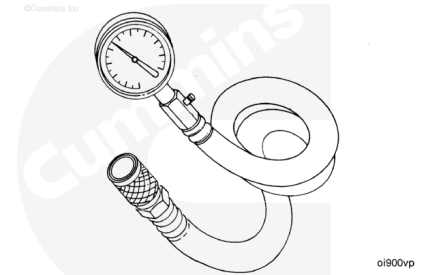


Trainings ▾

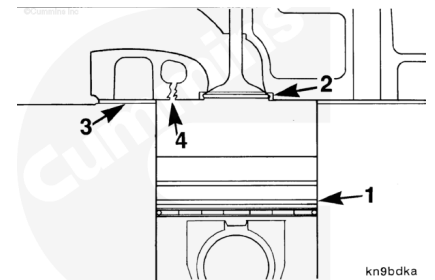
Measure

It is very time-consuming and expensive to begin removing internal engine components to diagnose failures. A compression gauge, Part Number 3164627, and adapter, Part Number 3164840, can be used as an aid to check for proper functioning of the following components:



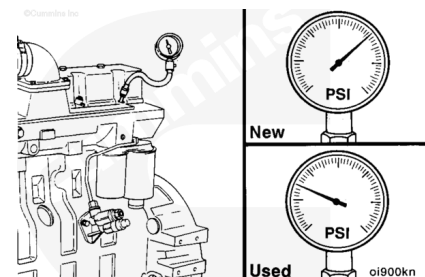
1. Piston ring sealing
2. Intake and exhaust valve sealing
3. Cylinder head gasket sealing
4. Cylinder head cracked

Refer to the appropriate procedure in this section for replacement of failed components.



Note : Due to variables such as starter and battery conditions that affect engine cranking speed, it is difficult to establish an absolute value for compression pressure; however, the following values can be used as guidelines:

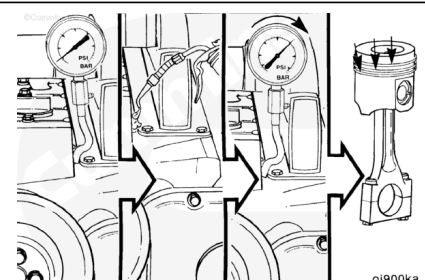
- New engine (cranking speed at 250 rpm) 2413 kPa [350 psi]
- Used engine (cranking speed at 250 rpm) 2068 kPa [300 psi]



It is recommended that the compression pressure be checked on all cylinders and then compared. All cylinders should be within 690 kPa [100 psi] of each other.

Piston Ring Sealing

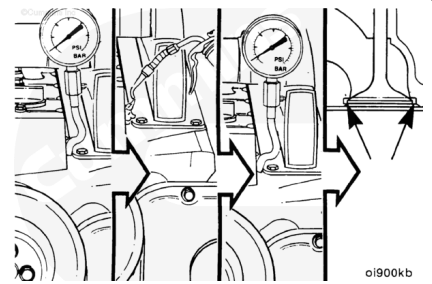
If the compression is low but can be increased significantly by squirting oil into the cylinder, the cause of low compression is inadequate sealing between the rings and the cylinder walls. Refer to procedure 007-013 for piston ring replacement.



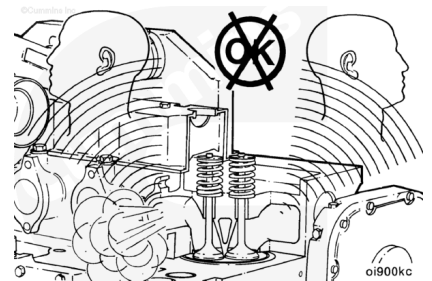
Intake and Exhaust Valve Sealing

If the compression is low on one or more non-adjacent cylinders and the pressure can **not** be increased by oiling the rings, poor valve sealing is to be suspected.

Refer to Procedure 007-009 for cylinder head replacement.



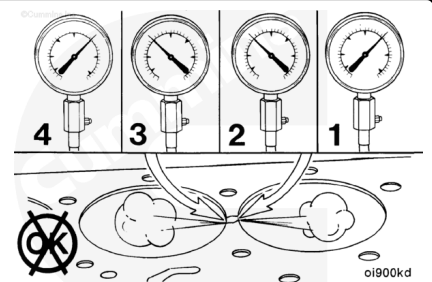
Valve leakage is often audible from the intake and exhaust manifold.



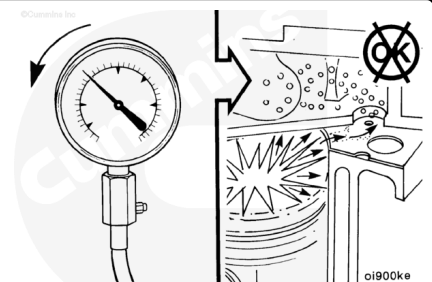
Cylinder Head Gasket Sealing

If the compression is low on adjacent cylinders and the pressure can **not** be increased by oiling the rings, the cylinder head gasket is probably leaking between the cylinders.

Refer to Procedure 007-009 for cylinder head gasket replacement.

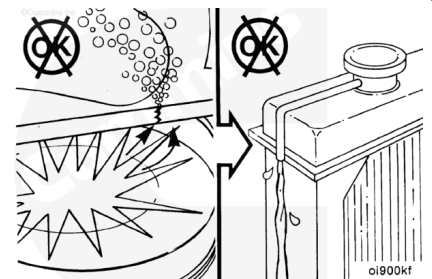


Note : Low compression on a single cylinder can be caused by an external leak to a coolant passage. A leak to a coolant passage of this magnitude will also result in coolant in the cylinder.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



A compression leak to the coolant normally will be detected by loss of coolant as the coolant is blown from the cooling system.

Note : Remove the drive belt from the water pump. Run the engine for 1 to 2 minutes and check for coolant being blown from the radiator by compression gases.